

init_chat_model

`init_chat_model` is a universal initializer function in LangChain that streamlines the configuration of chat models and reduces boilerplate — whether you're working with OpenAI's GPT, Anthropic's Claude, Cohere, or other providers.

It is used in two main scenarios:

- **Fixed Model**

```
from langchain.chat_models import init_chat_model

# Format: "provider:model_name"
gpt = init_chat_model("openai:gpt-4o", temperature=0)
claude = init_chat_model("anthropic:claude-3-5-sonnet-latest", temperature=0)
gemini = init_chat_model("google_vertexai:gemini-2.0-flash", temperature=0)

response = gpt.invoke("What is the capital of France?")
print(response.content)
```

- **Configurable Model (Switch at runtime)**

```
from langchain.chat_models import init_chat_model

# No model specified upfront – fully configurable
configurable_model = init_chat_model(
    configurable_fields=("model", "model_provider"),
    temperature=0
)

# Use GPT-4o
configurable_model.invoke(
    "Hello!",
    config={"configurable": {"model": "gpt-4o", "model_provider": "openai"}}
)

# Switch to Claude – same code, no changes!
configurable_model.invoke(
    "Hello!",
    config={"configurable": {"model": "claude-3-5-sonnet-latest", "model_provider":
"anthropic"}}
)
```

- **Configurable with Defaults + `config_prefix`**

```

configurable_model = init_chat_model(
    "openai:gpt-4o",
    configurable_fields="any",    # allow overriding ANY param at runtime
    config_prefix="foo",        # prefix for runtime config keys
    temperature=0,
)

# Use defaults (GPT-4o, temp=0)
configurable_model.invoke("What's your name?")

# Override model and temperature at runtime
configurable_model.invoke(
    "What's your name?",
    config={
        "configurable": {
            "foo_model": "anthropic:claude-opus-4-6",
            "foo_temperature": 0.7,
        }
    }
)

```

In the end, it is used to initialize any chat model in LangChain — especially useful when building apps that need to support multiple providers or switch models dynamically.